AI Agent Watchdog

Portfolio proof of concept | AI reliability, incident visibility, and workflow escalation

Inspired by a discussion about practical AI applications, this project explores a potential monitoring layer that can help identify, record, and route AI-agent output issues. It is a concept for workflow optimization, not a claim that a particular company or platform has a confirmed gap.

What I built

- **Python:** Created synthetic test cases and rule-based checks for incorrect answers and citation issues; logged incidents and assigned illustrative severity levels.
- **Incident log:** Exported structured incident records to CSV for use by other tools.
- **Power BI:** Built a dashboard to review incident counts, severity, test results, and incident details.
- **Google Drive + n8n:** Stored and read the CSV, filtered high-severity incidents, and routed them through an alert workflow.
- **Email notifications:** Tested email alerts that summarize available incident details and request human review.

Purpose and potential value

The purpose is to make potential AI failures easier to see and act on, rather than relying entirely on manual checks. The dashboard supports investigation, while automation can bring selected incidents to a reviewer's attention.

- **Earlier visibility:** Surface defined failure signals before they are overlooked in a large volume of outputs.
- **Consistent handling:** Apply the same agreed rules to every record in scope.
- **Traceability:** Keep structured incident records that can be reviewed and analyzed.
- **Targeted escalation:** Notify people about selected high-priority cases instead of sending every record.
- **Complementary design:** Could sit alongside existing AI and operational tools if a specific monitoring need is identified.
- **Potential to start small:** A pilot could focus on one workflow and expand only if results justify it.

What we could build next

- **Live test-agent integration:** Connect an authorized test agent and collect outputs in a controlled environment.
- **Stronger evaluation:** Add defined test suites, structured-output checks, reference validation, and human-reviewed quality labels.
- **Persistent incident tracking:** Store incident IDs and processing state in a database to support reliable cross-run deduplication and audit history.
- **Trend and root-cause assistance:** Identify recurring patterns and use an LLM to suggest possible causes, clearly marked for validation.
- **Controlled remediation:** Propose bounded fixes, test them in a sandbox, rerun failed checks, and record results.
- **Human approval and safeguards:** Require review before consequential changes reach live systems.

Why consider it—and where it fits

Watchdog is not positioned as something everyone should choose instead of Copilot, HCSS, or another existing platform. Those tools serve different purposes and may already offer relevant monitoring capabilities depending on product and configuration. The potential case for Watchdog is as a focused, configurable reliability layer when an organization needs additional checks, incident tracking, or escalation around AI outputs.

- **Consider it when:** AI outputs affect operational decisions, teams need explicit evaluation rules, failures must be logged, or high-priority exceptions need a defined review path.
- **Potential advantages:** Tailored rules, transparent incident records, customizable severity thresholds, and integration with existing reporting or notification workflows.
- **Adoption principle:** Compare the cost, coverage, security, integration effort, and measurable results against capabilities already available before choosing or expanding the solution.

Current scope and limitations

Built and tested: a prototype using synthetic incident data, a Power BI dashboard, and an n8n-to-email alert workflow. **Not yet built:** live production monitoring, LLM-based diagnosis, or automatic repair. The current checks are rule-based and do not guarantee that an arbitrary answer is correct. Cross-run duplicate suppression depends on the deduplication configuration and has not been established as production-grade persistent tracking.

What this demonstrates

Translating a business idea into a testable proof of concept; connecting Python data checks, CSV logging, Power BI reporting, and n8n automation; testing an end-to-end workflow; and distinguishing current functionality from future opportunities.